Commutative Algebra & Algebraic Geometry

Exercise 3

Ari Feiglin

3.1 Problem

(1) Prove that $\operatorname{Spec}(R)$ is connected iff its **reduction** $\operatorname{Spec}(R)_{\mathsf{red}} = \operatorname{Spec}(R/\operatorname{nil}(R))$ is connected.

(2) Compute the connected components of $X = \operatorname{Spec}(R)$ in the following cases:

- **(i)** $R = \mathbb{Z}/60\mathbb{Z}$,
- **(ii)** $R = \mathbb{C}[x,y,z]/(x^2+y^2-z^2, x^2+y^2-z)$.

(3) Find a polynomial $f \in \mathbb{R}[x,y]$ such that $\operatorname{Spec}(\mathbb{R}[x,y]/(f))$ is connected but $\operatorname{Spec}(\mathbb{C}[x,y]/(f))$ is disconnected.

(1) Consider the projection $\pi\colon R \to R/\operatorname{nil}(R)$. From the previous problem set, we know that this induces $\pi^*\colon \operatorname{Spec}(R/\operatorname{nil}(R)) \to \operatorname{Spec}(R)$, a topological embedding onto its image $V(\operatorname{nil}(R))$. Since $V(\operatorname{nil}(R)) = \operatorname{Spec}(R)$, we get that π^* is a homeomorphism $\operatorname{Spec}(R/\operatorname{nil}(R)) \to \operatorname{Spec}(R)$. Since connectedness is a topological property, the result follows.

(2)

(i) For $R = \mathbb{Z}/60\mathbb{Z}$ we note that the prime factorization of 60 is $60 = 2^2 \cdot 3 \cdot 5$ so that

$$R \cong (\mathbb{Z}/4\mathbb{Z}) \times (\mathbb{Z}/3\mathbb{Z}) \times (\mathbb{Z}/5\mathbb{Z})$$

so that

$$\operatorname{Spec}(R) \cong \operatorname{Spec}(\mathbb{Z}/4\mathbb{Z}) \sqcup \operatorname{Spec}(\mathbb{Z}/3\mathbb{Z}) \times \operatorname{Spec}(\mathbb{Z}/5\mathbb{Z}).$$

The ideals inducing the ring isomorphism are $4\mathbb{Z}/60\mathbb{Z}$, $3\mathbb{Z}/60\mathbb{Z}$, $5\mathbb{Z}/60\mathbb{Z}$, so that

$$\operatorname{Spec}(R) = V(4\mathbb{Z}/60\mathbb{Z}) \sqcup V(3\mathbb{Z}/60\mathbb{Z}) \sqcup V(5\mathbb{Z}/60\mathbb{Z}).$$

(Meaning for these coprime ideals I_1, I_2, I_3 we have $R \cong (R/I_1) \times (R/I_2) \times (R/I_3)$, and we saw in recitation this implies $\operatorname{Spec}(R) = V(I_1) \sqcup V(I_2) \sqcup V(I_3)$.) Each of these rings are connected (for $3, 5$ this is because they are fields, and $n^2 \equiv n \pmod 4$ is only solved by $0, 1$), so these are the connected components of $\operatorname{Spec}(R)$.

(ii) Note that

$$(x^2+y^2-z^2, x^2+y^2-z) = (x^2+y^2-z, z^2-z).$$

The right-hand side is contained in the left because z^2-z is equal to $(x^2+y^2-z)-(x^2+y^2-z^2)$. The left-hand side is contained in the right because $x^2+y^2-z^2 = (x^2+y^2-z)-(z^2-z)$.

Now we see that $(x^2+y^2-z, z), (x^2+y^2-z, z-1)$ are comaximal ($1 = z-(z-1)$), and their product is the given ideal $(x^2+y^2-z^2, z(z-1))$. Thus

$$R \cong \frac{\mathbb{C}[x,y,z]}{(x^2+y^2-z, z)} \times \frac{\mathbb{C}[x,y,z]}{(x^2+y^2-z, z-1)}$$

by the chinese remainder theorem. Now for $f \in \mathbb{C}[x,y]$, $\mathbb{C}[x,y,z]/(f-z, z-a) \cong \mathbb{C}[x,y]/(f-a)$. This is because $(f-z, z-a) = (f-a, z-a)$, and $\mathbb{C}[x,y,z]/(f-a) \to \mathbb{C}[x,y]/(f-a)$ given by mapping $g(x,y,z)+(f-a)$ to $g(x,y,a)+(f-a)$ is a well-defined surjective morphism. Its kernel is then $(f-a, z-a)/(f-a)$, and we get the desired isomorphism.

So we have that

$$R \cong \frac{\mathbb{C}[x,y]}{(x^2+y^2)} \times \frac{\mathbb{C}[x,y]}{(x^2+y^2-1)}.$$

These are both irreducible since $(x^2+y^2), (x^2+y^2-1)$ are both prime (the polynomials are irreducible), and thus the rings in the product are integral domains and this irreducible (since $\text{nil}(R) = 0$ for an integral domain, which is prime again since it's an integral domain). Thus they are also connected.

So the connected components of R are the vanishing sets of the ideals by which we quotiented. That is,

$$X = V\left(\frac{(x^2+y^2-z, z)}{(x^2+y^2-z^2, x^2+y^2-z)}\right) \sqcup V\left(\frac{(x^2+y^2-z, z-1)}{(x^2+y^2-z^2, x^2+y^2-z)}\right)$$

are the connected components of X.

(3) Let $f = x^2+1$, which is irreducible over $\mathbb{R}$ and thus $(f) \leq \mathbb{R}[x,y]$ is prime. Then $\mathbb{R}[x,y]/(f)$ is an integral domain and thus has no nontrivial idempotents (if $f(f-1) = 0$ in an integral domain, $f = 0$ or 1), so it is connected. But let $g = \frac{ix+1}{2} \in \mathbb{C}[x,y]$, then

$$g(g-1) = \frac{(ix+1)(ix-1)}{4} = \frac{-x^2-1}{4} = -\frac{x^2+1}{4},$$

which is zero modulo f. Thus g is a nontrivial idempotent in $\mathbb{C}[x,y]/(f)$ and thus it is disconnected.

3.2 Problem

Let X be a topological space.

(1) Show that the following are equivalent:

- **(i)** X is irreducible,
- **(ii)** Any nonempty open subset of X is dense,
- **(iii)** Any two nonempty open subsets have nonempty intersection.

(2) Let $Y \subseteq X$ be a subspace. Show the following.

- **(i)** Y is irreducible iff $\overline{Y}$ is irreducible.
- **(ii)** If Y is irreducible then it is contained in an irreducible component of X.

(3) Prove that X is a union of its irreducible components and they are closed.

(4) Assume X is Hausdorff, describe X's irreducible components.

(5) Now assume $X = \text{Spec}(R)$.

- **(i)** Show that an irreducible component of X is the same as $V(\mathfrak{p})$ where $\mathfrak{p} \in X$ is minimal.
- **(ii)** Compute the irreducible components of X in the case $R = \mathbb{C}[x,y]/(x^2)$ and $R = \mathbb{C}[x,y,z]/(x^2+y^2-1, x^2+z^2-1)$.

(1) A set is dense iff it intersects every nonempty open subset of X. Thus the second and third points are clearly equivalent. If X is irreducible, then let $U \subseteq X$ be nonempty open. It is dense because

$\overline{U} \cup (X - U) = X$ is a union of closed sets, and since X is irreducible one must be equal to X, so $\overline{U} = X$ meaning U is dense.

Conversely suppose every two nonempty open subsets have nonempty intersection, and $X = Z \cup W$ for closed sets Z, W. Then $Z^c \cap W^c = \emptyset$ for open subsets Z^c, W^c. Thus one must be empty, so its complement is all of X; X is then irreducible.

(2) If Y is reducible, then $Y = Z \cup W$ for proper closed subsets $Z, W \subseteq Y$. Then $\overline{Y} = \overline{Z} \cup \overline{W}$, and these are proper closed subsets: $Z = \overline{Z} \cap Y$ so if $\overline{Z} = \overline{Y}$ then $Z = \overline{Y} \cap Y = Y$, a contradiction. Thus $\overline{Y}$ is reducible.

And if $\overline{Y}$ is reducible, $\overline{Y} = Z \cup W$ for proper closed subsets, then $Y = (Z \cap Y) \cup (W \cap Y)$ is a union of closed sets. If $Z \cap Y = Y$ then $Y \subseteq Z$ so $\overline{Y} \subseteq \overline{Z} = Z$ a contradiction (since $\overline{Y}$ is closed, relatively closed subsets are closed). Thus Y is reducible. So we have shown the first point.

Let $\mathcal{X}$ be the set of all irreducible subsets of X containing Y, it is nonempty since $Y \in \mathcal{X}$. Then we apply Zorn to $\mathcal{X}$ to obtain a maximal irreducible component. Let $C \subseteq \mathcal{X}$ be a chain of irreducible subsets of X containing Y, we claim its union $\mathcal{Y} = \bigcup C$ is irreducible. If $\mathcal{Y} = Z \cup W$ is a union of closed subsets and say $W \neq \mathcal{Y}$ so let $x \in \mathcal{Y} - W$; we claim $Z = \mathcal{Y}$. Let $y \in \mathcal{Y}$, then there exists a $Y' \in C$ such that $x, y \in Y'$ (since C is a chain), and $Y' = (Z \cap Y') \cup (W \cap Y')$. Since Y' is irreducible, either $Z \cap Y' = Y'$ or $W \cap Y' = Y'$. But $x \notin W \cap Y'$ so $W \cap Y' \neq Y'$ and thus $Z \cap Y' = Y'$, meaning $y \in Z$. Thus $Z = \mathcal{Y}$ as desired.

So $\mathcal{Y}$ is irreducible and maximal by definition.

(3) Each irreducible component $\mathcal{Y}$ is closed because $\overline{\mathcal{Y}}$ is irreducible and contains $\mathcal{Y}$, so by maximality $\mathcal{Y} = \overline{\mathcal{Y}}$. And for every $x \in X$, $\{x\}$ is irreducible and thus contained in an irreducible component. So the union of all irreducible components contains every point in X, as desired.

(4) If X is Hausdorff, its irreducible components are the singletons. If $Y \subseteq X$ has at least two points $x \neq y \in Y$ then they have disjoint neighborhoods $x \in U, y \in V$. Since $U \cap V = \emptyset$, by the first point Y cannot be irreducible. And singletons are clearly irreducible, thus proving the claim.

(5)

(i) The closed sets of X are of the form $V(J)$ for an ideal $J \leq R$. An irreducible component is closed, thus of this form. Now for prime $\mathfrak{p}$, $V(\mathfrak{p})$ is irreducible: if $V(\mathfrak{p}) = V(I) \cup V(J) = V(IJ)$ then $IJ \leq \mathfrak{p}$, so by primality either $I \leq \mathfrak{p}$ or $J \leq \mathfrak{p}$, and then $V(\mathfrak{p}) \subseteq V(I)$ or $V(J)$, thus $V(\mathfrak{p})$ is indeed irreducible.

Since $V(J) = V(\sqrt{J})$ we focus on radical ideals. We claim that $V(J)$ is reducible for a radical ideal J if it is not prime. If J is not prime, there are ideals I_1, I_2 such that $I_1 I_2 \leq J$ but I_1, I_2 are not contained in J. Then $V(J) \subseteq V(I_1 I_2)$, so

$$V(J) = V(I_1 I_2) \cap V(J) = (V(I_1) \cup V(I_2)) \cap V(J) = V(I_1 + J) \cup V(I_2 + J).$$

We claim that for $I = I_1, I_2$, $V(I + J) \neq V(J)$. Indeed, this equality holds only when $J = \sqrt{I + J}$, but $I \leq \sqrt{I + J}$ and this is not contained in J, so $V(J) \neq V(J + I)$ so that $V(J)$ is reducible.

So all irreducible subsets are of the form $V(\mathfrak{p})$ for $\mathfrak{p}$ prime. And for prime ideals $\mathfrak{p}, \mathfrak{q}$, $V(\mathfrak{p}) \subseteq V(\mathfrak{q})$ iff $\mathfrak{q} \leq \mathfrak{p}$ (since $\mathfrak{p} \in V(\mathfrak{q})$ implies $\mathfrak{q} \leq \mathfrak{p}$ and in the other direction V is order-reversing). Thus a maximal irreducible subset is a minimal prime ideal, as desired.

(ii) For $R = \mathbb{C}[x, y]/(x^2)$ we recall that prime ideals of R are prime ideals of $\mathbb{C}[x, y]$ containing (x^2). Prime ideals of $\mathbb{C}[x, y]$ are of the form $(f(x, y))$ for f irreducible or $(x - a, y - b)$ (maximal ideals). The latter contains (x^2) only when $a = 0$. For the former, $(x^2) \subseteq (f(x, y))$ iff f divides x^2, so $f = x$.

Thus the prime ideals of R are $(x)/(x^2)$ and $(x, y - b)/(x^2)$. The only minimal prime ideal is $(x)/(x^2)$, it is a minimum prime ideal, thus $V((x)/(x^2)) = X$ is irreducible. Alternatively, we note that $\sqrt{(x^2)} = (x)$ is prime, so $\mathrm{nil}(R)$ is prime and thus X is irreducible.

For $R = \mathbb{C}[x, y, z]/(x^2 + y^2 - 1, x^2 + z^2 - 1)$. Let I be the ideal, then notice that

$$(x^2 + y^2 - 1) - (x^2 + z^2 - 1) = y^2 - z^2 = (y - z)(y + z) \in I.$$

Now we claim that $I+(y-z), I+(y+z)$ are the minimal prime ideals containing I. Note that $I+(y-z) = (x^2+y^2-1, y-z)$ (since modulo $z-y$, $x^2+z^2-1 = x^2+y^2-1$), and $I+(y+z) = (x^2+y^2-1, y+z)$. Then

$$\frac{\mathbb{C}[x,y,z]}{(x^2+y^2-1, y-z)} \cong \frac{\mathbb{C}[x,y]}{(x^2+y^2-1)}$$

by mapping $f(x,y,z)$ to $f(x,y,y)$ in $\mathbb{C}[x,y,z]/(x^2+y^2-1)$. Similarly

$$\frac{\mathbb{C}[x,y,z]}{(x^2+y^2-1, y+z)} \cong \frac{\mathbb{C}[x,y]}{(x^2+y^2-1)}.$$

Since x^2+y^2-1 is irreducible over $\mathbb{C}[x,y]$, it generates a prime ideal and thus $I+(y-z), I+(y+z)$ are prime as well.

Now if $I \leq \mathfrak{p}$ for any other prime ideal, then $(y-z)(y+z) \in I$ so $y-z$ or $y+z$ is in $\mathfrak{p}$. Thus $I+(y-z)$ or $I+(y+z)$ is contained in $\mathfrak{p}$, so these are the minimal prime ideals containing I. So $(I+(y\pm z))/I$ are minimal prime ideals and thus their vanishing sets are the irreducible components of X.

3.3 Problem

Let R be a PID, and M a fg R-module.

(1) Show that R is Noetherian.

(2) Now show that if $M \leq R^n$, then M is free of rank $\leq n$.

(3) Find $n, k \in \mathbb{N}$ and a matrix $C \in M_{n\times k}(R)$ such that $M \cong R^n/\mathrm{colsp}(C)$.

(4) Let $D_1 \in \mathrm{GL}_n(R)$ and $D_2 \in \mathrm{GL}_k(R)$ be invertible. Show that

$$R^n/\mathrm{colsp}(D_1C) \cong M \cong R^n/\mathrm{colsp}(CD_2).$$

This allows us to do row/column operations on C without changing M.

(5) Assume $c_{1,1} \nmid c_{1,2}$. Show that $(c_{1,1}, c_{1,2}) = (d)$ for some $d \in R$ and that there exist $s,t,x,y \in R$ such that $d = sc_{1,1} + tc_{1,2}$ and $c_{1,1} = xd$ and $c_{1,2} = yd$. Furthermore show that

$$D = \begin{pmatrix} s & -y & \\ t & x & \\ & & I_{k-2} \end{pmatrix}$$

is invertible, and $C' = CD$ has $c'_{1,1} = d$.

(6) If $c_{1,1} \nmid c_{1,j}$ for some j, swap $c_{1,2}$ with $c_{1,j}$ (via an elementary column operation, so it doesn't change M), and repeat the previous point. Show that in each step, $(c_{1,1}) \subsetneq (c'_{1,1})$ and explain why this means the process terminates.

(7) Explain why we may replace C by a matrix such that its first column and row are zero except for $c_{1,1}$.

(8) Show that C can be replaced by a matrix which is nonzero outside of its diagonal and $c_{1,1} \mid c_{2,2} \mid \cdots$.

(9) Conclude the invariant factor decomposition theorem from recitation.

(1) R is Notherian because every ideal is finitely generated (alternatively the union of any ascending chain of ideals is an ideal, principal because R is a PID, and its generator must lie in one of the ideals and thus it stabilizes at that ideal).

(2) For $n = 1$, if $M \leq R$ then M is an ideal, thus of the form $M = (a)$ for $a \in R$ since it is a PID. If $a \neq 0$ then $\{a\}$ forms a basis for M and thus it is of rank 1, otherwise $M = 0$ and it is free of rank zero.

Inductively, let $M' = M \cap (0 \times R^{n-1})$ so that $M' \leq R^{n-1}$ and thus is free of rank $\leq n-1$. Let $\pi_1 \colon R^n \to R$ project onto the first coordinate, then this is a morphism of R-modules and thus $\pi_1(M) \leq R$ is a submodule. By our base case, $\pi_1(M) = (f)$ for some $f \in R$.

Now let $m \in M$ such that $\pi_1(m) = f$. Let B' be a basis for M' of cardinality $\leq n-1$, and define B to be B' if $f = 0$ and $B' \cup \{m\}$ otherwise. We claim that B is a basis for M. It is linearly independent since if $\sum_i a_i m_i' + am = 0$ for $a_i, a \in R$ and $m_i' \in B'$, then projecting onto the first coordinate gives

$$0 = \pi_1\left(\sum_i a_i m_i' + am\right) = af$$

and so $a = 0$ since R is an integral domain. By B''s linear independence, $a_i = 0$ for all i as well, so B is linearly independent (for $f = 0$ it is linearly independent as it is equal to B'). Now for $x \in M$, we have that $\pi_1(x) = af$ and thus $x - af \in B'$ so that B spans M, so it is a basis.

Thus M has a basis of cardinality $\leq n$, as desired.

(3) Suppose that M is generated by n elements, let $B = (m_1, \ldots, m_n)$ generate it. Then $\pi \colon R^n \to M$ by $e_i \mapsto m_i$ is a surjective map of R-modules, and thus $R^n/\ker\pi \cong M$. Since $\ker\pi \leq R^n$, by the above point it is free with rank $k \leq n$. Let $(v_1, \ldots, v_k)$ be a basis for $\ker\pi$ in R^n, then its span is just the column-span of the matrix

$$C = \begin{pmatrix} | & & | \\ v_1 & \cdots & v_k \\ | & & | \end{pmatrix},$$

which is of size $n \times k$ as desired.

(4) Note that $\operatorname{colsp}(D_1C) = D_1\operatorname{colsp}(C)$, since the columns of D_1C are the columns of C times D_1. Define $\phi \colon R^n \to R^n/\operatorname{colsp}(D_1C)$ by $v \mapsto D_1v + \operatorname{colsp}(D_1C)$. Since D_1 is invertible, this is surjective, and $D_1v \in \operatorname{colsp}(D_1C) = D_1\operatorname{colsp}(C)$ iff $v \in \operatorname{colsp}(C)$ because D_1 is invertible. Thus its kernel is $\operatorname{colsp}(C)$ so that it forms an isomorphism $M \cong R^n/\operatorname{colsp}(C) \cong R^n/\operatorname{colsp}(D_1C)$.

And notice that $\operatorname{colsp}(CD_2) = C\operatorname{colsp}(D_2) = C \cdot R^k = \operatorname{colsp}(C)$, since invertible matrices have full column span.

Since elementary row/column matrices are invertible, we see that we can apply row/column operations to C without changing M.

(5) Since R is a PID, $(c_{1,1}, c_{1,2}) = (d)$ for some $d \in R$ since its an ideal. Then $d = sc_{1,1} + tc_{1,2}$ since $d \in (c_{1,1}, c_{1,2})$, and since $c_{1,1}, c_{1,2} \in (d)$ we have $c_{1,1} = xd$ and $c_{1,2} = yd$.

Matrices are invertible iff their determinant is a unit (via the adjugate matrix). D's determinant is $sx + ty$. We know that

$$d = sc_{1,1} + tc_{1,2} = sxd + tyd \Longrightarrow sx + ty = 1$$

because this is an integral domain. So D is indeed invertible (its inverse is $(x, y;\ -t, s) \oplus I_{k-2}$).

And

$$c_{1,1}' = \sum_{i=1}^{k} c_{1,i}d_{i,1} = c_{1,1}d_{1,1} + c_{1,2}d_{2,1} = sc_{1,1} + tc_{1,2} = d,$$

and

$$c_{1,2}' = \sum_{i=1}^{k} c_{1,i}d_{i,2} = -yc_{1,1} + xc_{1,2},$$

which is divisible by d.

(6) If $c_{1,1} \nmid c_{1,j}$ then via an elementary column operation $C_2 \leftrightarrow C_j$, we repeat the previous step, and then swap C_2 and C_j back. In each step we have that $(c_{1,1}) \subseteq (c'_{1,1})$ because $c'_{1,1}$ is the generator of the ideal $(c_{1,1}, c_{1,j})$ which contains $(c_{1,1})$. Since R is Notherian, this ascending chain of ideals must stabilize, so this process terminates.

At each step we have $c_{1,j} \in (c'_{1,1})$ so that at the end of the process, we have $c_{1,j} \in (c'_{1,1})$ for all j meaning that at the end $c_{1,1}$ divides every element in its row. If we do the same thing to its column, we get that $c_{1,1}$ divides every element in its column. Since we just shrunk $c_{1,1}$ (relative to the divisibility preorder), $c_{1,1}$ still divides its row.

(7) We can then perform elementary row and column operations to make all elements in the first row and column other than $c_{1,1}$ zero. Explicitly, for $c_{1,j}$ say, since $c_{1,1}$ divides it, there is an $a \in R$ such that $c_{1,j} = ac_{1,1}$. Then the elementary column operation $C_j \leftarrow C_j - aC_1$ makes $c_{1,j}$ zero.

(8) We can repeat this process on the submatrix of C starting at $c_{2,2}$, and then at $c_{3,3}$, and so on. In the end we end up with a matrix of the form

$$C = \begin{pmatrix} c_{1,1} & & \\ & c_{2,2} & \\ & & \ddots \end{pmatrix}.$$

Now we add the second row to the first (an elementary row operation), and let $d = (c_{1,1}, c_{2,2})$ so there are $s, t, x, y \in R$ such that $d = sc_{1,1} + tc_{2,2}$ and $c_{1,1} = xd$ and $c_{2,2} = yd$ as before. Then the same matrix D from the fifth point is invertible and transforms C only in the top left submatrix to

$$\begin{pmatrix} s & -y \\ t & x \end{pmatrix} \begin{pmatrix} c_{1,1} & c_{2,2} \\ & c_{2,2} \end{pmatrix} = \begin{pmatrix} d & -yc_{1,1} + xc_{2,2} \\ tc_{2,2} & xc_{2,2} \end{pmatrix},$$

all of its components are all divisible by d so after row and column operations we get a matrix where $c_{1,1} = d$ and $c_{2,2}$ is divisible by $c_{1,1}$.

We repeat this until $c_{1,1}$ divides all $c_{i,i}$, which must terminate as $(c_{1,1}) \subseteq (c'_{1,1})$. Now we do this same process for $c_{2,2}$ and $c_{i,i}$ for $i > 2$ Since $c_{1,1}$ divides $c_{2,2}$ and $c_{i,i}$ the resulting $c_{2,2}$ and $c_{i,i}$ will still be divisible by $c_{1,1}$ (since $(c_{2,2}, c_{i,i}) \subseteq (c_{1,1})$). We repeat this for the entire diagonal, going downward, giving us the desired result.

(9) So we have that $M \cong R^n/\text{colsp}(C)$ where C is diagonal (not square) with $c_1 \mid c_2 \mid \cdots$. Then we have that

$$\text{colsp}(C) = \text{colsp} \begin{pmatrix} c_1 & & \\ & c_2 & \\ & & \ddots \end{pmatrix} = \text{span}\left\{c_i e_i \mid 1 \leq i \leq \ell\right\},$$

where $e_i \in R^n$ are the standard basis vectors, and c_i are nonzero diagonal elements. Define $\phi\colon R^n \to R^{n-\ell} \oplus \bigoplus_{i=1}^{\ell} R/(c_i)$ by

- **(i)** for $1 \leq i \leq \ell$, map $e_i \in R^n$ to $1 + (c_i) \in R/(c_i)$,
- **(ii)** for $\ell < j \leq n$, map $e_j \in R^n$ to $e_{j-\ell} \in R^{n-\ell}$.

This is surjective, since for $(v_1, \ldots, v_{n-\ell}) \in \mathbb{R}^{n-\ell}$ and $f_i + (c_i) \in R/(c_i)$, ϕ maps $\sum_{i=1}^{\ell} f_i e_i + \sum_{j=\ell+1}^{n} v_{j-\ell} e_j$ to this. Now, $(\overline{v}, \overline{f} + (\overline{c}))$ is in ϕ's kernel iff $\overline{v} = 0$ and $f_i \in (c_i)$. Thus

$$\ker\phi = \left\{ \sum_{i=1}^{\ell} f_i e_i \;\middle|\; f_i \in (c_i) \right\} = \text{span}\left\{c_i e_i \mid 1 \leq i \leq \ell\right\} = \text{colsp}(C).$$

So ϕ induces an isomorphism

$$M \cong R^n/\text{colsp}(C) \cong R^{n-\ell} \oplus \bigoplus_{i=1}^{\ell} R/(c_i),$$

and $c_1 \mid c_2 \mid \cdots$ as desired.

3.4 Problem

Assume R is a domain and let M be a fg R-module.

(1) In class we showed that if M is free, it is torsion-free.

(i) Show that if R is a PID and M is torsion-free, it is free.

(ii) Find an example where R is not a PID and M is torsion-free but not free.

(2) Let M_{tor} be the subset of torsion elements of M. Show that:

(i) M_{tor} is a submodule of M,

(ii) M/M_{tor} is torsion-free,

(iii) For $S = R - 0$, let $\iota\colon M \to S^{-1}M$ be the structure morphism, then $\ker\iota = M_{\text{tor}}$.

(1)

(i) We construct a maximal linearly independent set B inductively. Start with some $0 \neq b_1 \in M$, and assuming we have $b_1, \ldots, b_n$ if there exists a $b_{n+1} \in M$ such that $b_1, \ldots, b_{n+1}$ is linearly independent, add it to B. Since $b_1, \ldots, b_{n+1}$ is linearly independent, the submodule generated by $b_1, \ldots, b_n$ is strictly contained in that generated by $b_1, \ldots, b_{n+1}$. Since R is a PID and M is fg, M is Noetherian, so this procedure must terminate. Thus we have a finite maximal linearly independent set $B = \{b_1, \ldots, b_k\}$.

Consider a generating set $m_1, \ldots, m_n \in M$. For each i, $B \cup \{m_i\}$ is dependent so $a_i m_i \in \operatorname{span}(B)$ for some $0 \neq a_i \in R$. If we let $a = a_1 \cdots a_n$ then $am_i \in \operatorname{span}(B)$ for all i, and thus $aM \subseteq \operatorname{span}(B)$. Since R is a domain and $a_i \neq 0$, a is nonzero as well.

Since B is finite and free, by the previous question submodules of a submodule over a PID are free, so aM is free. Multiplication by a, $M \to aM$, is an isomorphism since $a \neq 0$ and M is torsion-free so $am \neq 0$ for $m \neq 0$, and this is clearly surjective. Thus $M \cong aM$ which is free, so M is free.

(ii) Let R be a domain, and $I \leq R$ an ideal. Then I is free iff it is principal. Clearly if $I = (a) = Ra$ then multiplication by a gives an isomorphism of modules $R \to I$, so that I is free. If I is free, then let B be a basis. If it has two elements $f \neq g \in B$ then $fg - gf = 0$ which contradicts linear independence. Thus B is a singleton $\{a\}$, and thus $I = Ra = (a)$ is principal.

Since R is a domain, $I \leq R$ is always torsion-free.

So for example take $R = \mathbb{C}[x, y]$ and the maximal ideal $I = (x, y)$. I is not principal, since if $(x, y) = (f)$ then x, y both divide f and thus $f = xyg$ for some polynomial g (by primality of x, y and x not dividing y). But then $x, y \notin (f)$ a contradiction. So we see that I is torsion-free (R is a domain), but not free.

(2)

(i) Clearly $0 \in M_{\text{tor}}$ since it is torsion. If $m \in M_{\text{tor}}$ is torsion, say $am = 0$, and $b \in R$, then $a(bm) = b(am) = 0$ so that $bm \in M_{\text{tor}}$. And if $m, n \in M_{\text{tor}}$ are torsion, so say $am = bn = 0$, then $ab(m + n) = 0$ and $ab \neq 0$ since R is a domain so that $m + n \in M_{\text{tor}}$. Thus M_{tor} is a submodule.

(ii) If $\overline{m} \in M/M_{\text{tor}}$ is torsion, then for some $0 \neq a \in R$, $am \in M_{\text{tor}}$. Then there exists a $0 \neq b \in R$ such that $bam = 0$, and since $ba \neq 0$ as R is a domain, $m \in M_{\text{tor}}$ so that $\overline{m} = 0$. Thus M/M_{tor} is torsion-free.

(iii) We have that $m \in \ker \iota$ iff $m/1 = 0$ iff there is an $a \in S$ such that $am = 0$. This just means that there is a $0 \neq a \in R$ such that $am = 0$, equivalently m is torsion. So $\ker \iota = M_{\text{tor}}$ as desired.